

Mathematics for Computing

Assignment 1 : Set Theory

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Assignment No.1

Questions

- 10 sets and their set builder form
- Example for every type of set from AI and ML Domain.
- General form of inclusion - exclusion principle
- Hasse Diagram of two partially ordered sets.

Introduction

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- Example of set: $\{a,e,i,o,u\}$
- Example of a set: Empty set (no items): $\{ \}$ or ϕ

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- Example of set: $\{1,7,19,20,14\}$
- Example of set: $\{a,e,i,o,u\}$
- Example of a set: Empty set (no items): $\{ \}$ or $\emptyset$
- Example of **not** a set: Set of all tall people in this room
- Example of **not** a set: The collection of the best football players in the world.

Set Builder Form

1. $A = \{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots\}$

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Set Builder Form:

$$Q_1 = \{(x, y) \mid x, y \in \mathbb{R} \text{ and } x > 0 \text{ and } y > 0\}$$

Set Builder Form

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$$E = \{x \mid x = 4y + 3, \forall y \in \mathbb{N}\}$$

4. Write a set of all integers whose square is strictly less than 50.

$$S = \{x \mid x^2 < 50 \forall x \in \mathbb{Z}\}$$

Set Builder Form

5. Translate the half open real interval $[-7, \infty)$ into formal set builder notation.

$$\mathbf{S} = \{x \mid -7 \leq x \leq \infty, \forall x \in \mathbb{R}\}$$

6. Write the set D of all positive integer divisor of 24.

$$\mathbf{D} = \{x \in \mathbb{Z} \mid kx = 24 \forall k \in \mathbb{Z}\}$$

Set Builder Form

7. Define the set **T** of all pythagorean triplets.

$$\mathbf{T} = \{x, y, z \mid \sqrt{x^2 + y^2} = z, \forall x, y, z \in \mathbb{Z}^+\}$$

8. Define set **M** of all natural numbers that are multiple of both 3 and 5.

$$\mathbf{M} = \{x \mid x = 3k, \forall k \in \mathbb{N} \text{ and } x = 5q, \forall q \in \mathbb{N}\}$$

Set Builder Form

9. Express the set **S** = {1, 8, 27, 64, 125, ...}

$$S = \{x \mid x = y^3 \ \forall y \in \mathbb{N}\}$$

10. Define the set **M** of all real numbers that are within a distance of 5 units from the number 12 on the number line.

$$\mathbf{M} = \{x : |x - 12| \leq 5\}$$

Example for every type of set from AI and ML Domain

- **Universal Set ($\mathcal{U}$):** The set of all elements under consideration, meaning any set A is a subset of it ($A \subseteq \mathcal{U}$).
ML Example: All possible emails that a spam filter could ever receive in the real world.
- **Finite Set:** A set with a countable, limited number of elements, where its size (cardinality) is $|S| = n$ for some integer n .
ML Example: The target labels in a sentiment analysis model: {Positive, Neutral, Negative}.

Example for every type of set from AI and ML Domain

- **Infinite Set:** A set with an endless number of elements, where $|S| = \infty$.

ML Example: All the possible decimal values a neural network weight can take between 0 and 1, represented mathematically as the continuous interval $[0, 1]$.

- **Empty Set ($\emptyset$):** A set containing zero elements, meaning $|\emptyset| = 0$.

ML Example: The overlap (intersection) between a properly split Training set (A) and Testing set (B). To avoid leakage, they share no elements: $A \cap B = \emptyset$.

Example for every type of set from AI and ML Domain

- **Singleton Set:** A set containing exactly one element, meaning $|S| = 1$, or $S = \{x\}$.
ML Example: A batch of data containing exactly one sample, used when a model is making a single real-time prediction in production.
- **Subset ($A \subseteq B$):** Every element in A is also contained in B , defined logically as $\forall x(x \in A \implies x \in B)$.
ML Example: The Validation dataset is a subset of the total collected historical data.

Example for every type of set from AI and ML Domain

- **Disjoint Sets:** Two sets that share no common elements whatsoever, defined as $A \cap B = \emptyset$.
ML Example: The set of images labeled "Cat" (A) and the set labeled "Car" (B), assuming an image can only be exactly one of these.

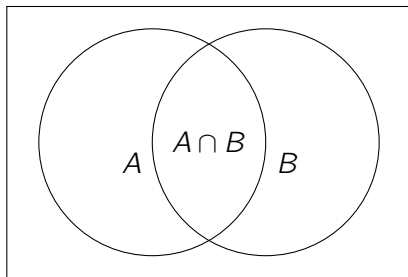
Inclusion - Exclusion Principle

- **What is Inclusion Exclusion Principle?**

The Inclusion Exclusion principle is a counting method used to find the number of elements in the union of multiple sets.

- For two sets **A and B**

$$|A \cup B| = |A| + |B| - |A \cap B|$$

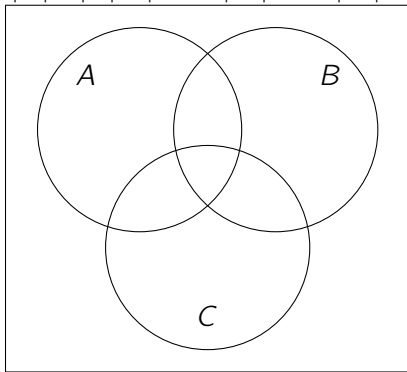


$$|A \cup B|$$

Inclusion - Exclusion Principle

- For three sets **A**, **B** and **C**

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$



$$|A \cup B \cup C|$$

General form of Inclusion-Exclusion Principle

For finite sets $A_1, A_2, A_3, \dots, A_n$, the total number of elements in their union is:

$$\begin{aligned} \left| \bigcup_{i=1}^n A_i \right| &= \sum_{i=1}^n |A_i| \\ &\quad - \sum_{1 \leq i < j \leq n} |A_i \cap A_j| \\ &\quad + \sum_{1 \leq i < j < k \leq n} |A_i \cap A_j \cap A_k| \\ &\quad - \dots \\ &\quad + (-1)^{n-1} |A_1 \cap A_2 \cap \dots \cap A_n| \end{aligned}$$

General form of Inclusion-Exclusion Principle

For finite sets $A_1, A_2, A_3, \dots, A_n$, the total number of elements in their union is:

$$\left| \bigcup_{i=1}^n A_i \right| = \sum_{k=1}^n (-1)^{k-1} \left(\sum_{1 \leq i_1 < i_2 < \dots < i_k \leq n} |A_{i_1} \cap A_{i_2} \cap \dots \cap A_{i_k}| \right)$$

What is a Poset?

A partially ordered set is a set where we have defined a specific rule for how elements in a set relate to each other.

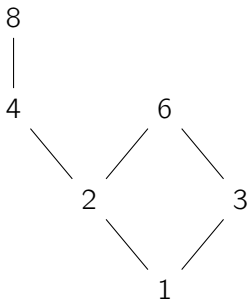
For a set to be a poset, the following three conditions must be met

- Reflexive: Every item relates to itself.
- Antisymmetric: If item **A** relates to item **B** and item **B** relates back to item **A**, then they must actually be the exact same item.
- Transitive: If **A** relates to **B** and **B** relates to **C**, then **A** automatically relates to **C**

Hasse Diagram

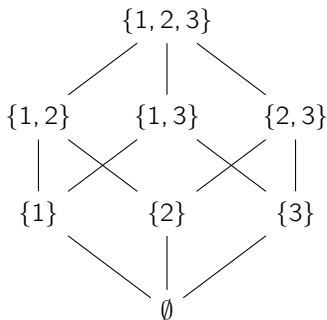
A graphical representation of a partial order relation in which all arrowheads are understood to be pointing upward is known as the Hasse Diagram.

Example: Let $A = \{1, 2, 3, 4, 6, 8\}$ be ordered by the relation “ a divides b ”.



Hasse Diagram

Example: The power set of $S = \{1, 2, 3\}$ ordered by subset inclusion ($\subseteq$).



Thank You

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